Question Bank

UIT	Branch- CSE/AIML/DS	Even Semester 2025-26		Semester: 4^{th}
	Subject: Operating System			Subject Code: BCS401
Unit –1: (Operating system and functions)				CO-1
Section-A (Short answer type questions)			Blooms Taxonomy Level	Session
1	A	What is spooling.	K1	[2014-15], [2015-16]
	B	Write a brief multiprocessor scheduling	K1	[2015-16]
	C	Define multithreading.	K1	[2015-16]
	D	What do you mean by kernel.	K1	[2015-16]
	E	Explain thread.	K2	[2016-17]
	F	Define Operating system and mention its major functions.	K1	[2021-22]
	G	Briefly define the term Real Time Operating System.	K2	[2021-22]
	H	Define two main functions of an operating system.	K1	[2022-23]
	I	Explain the principal advantages of multiprogramming.	K2	[2022-23]
	J	What is the primary function of an operating system?		2023-24, 2024-25
	K	Define a batch processing system?		2024-25
	L	A system has 3 processors and 5 programs ready for execution. In how many ways can the processors be assigned to the programs assuming each processor executes one program at a time?		2024-25
Section-B (Long answer type questions)				
2	A	Describe the differences between symmetric and asymmetric multiprocessing	K2	[2014-15]
	B	Discuss the various operating system components.	K2	[2014-15], [2017-18]
	C	Explain the layered architecture of Operating system also explain the advantage and disadvantage of layered design	K2	[2014-15] [2017-18]
	D	What do you mean by kernel. Describe monolithic and microkernel structure of operating system.	K2	[2018-19], [2016-17], [2021-22]
	E	Write down different type of operating system	K1	[2016-17]
	F	Define operating system. list the objective of operating system.	K1	[2017-18]
	G	Explain the batch operating system with example.	K2	[2018-19]
	H	Define operating system. Describe the operating system function.	K1	[2018-19]
	I	Explain in detail about the Multiuser Systems and Multithreaded Systems.	K2	[2021-22]
	J	Explain in detail about the Operating System services.	K2	[2021-22]
	K	Explain in detail about the Threads and their management.	K2	[2021-22]
	L	Explain the following terms in detail: (i) Multiprocessor operating system (ii) Real time system (iii) Time sharing system	K2	[2022-23]
	M	Compare and contrast time-sharing and multiprogramming operating system concepts?		2023-24
	N	How does a multiuser operating system differ from a multiprocessing system?		2023-24
	O	Explain the structure and components of an operating system with a layered architecture?		2023-24
	P	Discuss the differences between Batch, Interactive, and Time Sharing Operating Systems with real-world applications.		2024-25

Question Bank

UIT	Branch-CSE/AIML/DS	Even Semester 2025-26	Semester: 4th	
	Subject: Operating System		Subject Code: BCS401	
Unit – 2 (Concurrent Process)			CO-2	
Section-A (Short answer type questions)			Blooms Taxonomy Level	Session
1	A	What is the use of inter process communication and context switching?	K2	2017-2018
	B	Define Busy Waiting? How to overcome busy waiting using Semaphore operations.	K1	2017-18, 2015-16, [2022-23]
	C	Difference between Process and Program.	K2	2016-2017
	D	What is Critical Section?	K1	2016-2017
	E	What do you understand by critical section?	K2	2015-2016
	F	Define process.	K1	2015-2016
	G	What is busy waiting?	K1	2015-2016
	H	What are Semaphores?	K1	2015-16, 2022-23
	I	What do you mean by Concurrent Processes?	K2	[2021-22]
	J	Explain starvation problem and its solution	K2	[2022-23]
	K	Define the principle of concurrency?		2023-24
	L	What is the difference between a process and a program?		2023-24
	M	In a Producer-Consumer system, the buffer size is 4. If Producer produces 6 items and Consumer consumes 3, find the number of items in the buffer at the end.		2024-25
	N	A process executes the following code: fork(); fork(); fork(); How many processes are created in total, including the original?		2024-25
Section-B (Long answer type questions)				
2	A	Give the principles, mutual exclusion in critical section problem. Also discuss how well these principles are followed in Dekker's solution.	K2	2016-2017, 2024-25
	B	State the Producer-Consumer problem. Given a solution to the solution using semaphores.	K3	2016-2017, [2018-19], [2022-23
	C	Give the solution of Readers-Writers problem by using the concept of semaphore?	K3	2015-2016
	D	Define Message passing and shared memory inter process communication.	K1	2015-2016
	E	Write and explain Peterson solution to the critical section problem	K3	2015-2016
	F	Explain the need of process synchronization.	K2	2015-2016
	G	Explain the principle of concurrency.	K2	2014-2015
	H	What are Semaphores? What is the usage of Semaphores? Explain with suitable example.	K2	2014-2015
	I	Give the constraints given by Dijkstra that are imposed on any solution for critical section problem. Also give Dekker's solution for two process and check whether it satisfies the above constraints	K3	2014-2015
	J	Give reader/writer problem and its solution using semaphores.	K3	2014-2015
	K	Explain the need of process synchronization. How can the interprocess communication be achieved?	K4	2014-2015
	L	Discuss the characteristics of a critical section problem.	K2	2014-2015
	M	What is semaphore? Differentiate between binary and counting semaphore.	K2	2014-2015
	N	What is a Critical Section problem? Give the conditions that a solution to the critical section problem must satisfy.	K2	2017-18, 2018-19, 2024-25
	O	Explain what semaphores are, their usages, implementation given to avoid busy waiting and binary semaphores.	K2	2018-2019

	P	What is Dining Philosophers problem? Discuss the solution to Dining philosopher's problem using monitors.	K3	[2017-18], [2022-23], [2021-22]
	Q	Discuss Mutual-exclusion implementation with test and set() instruction.	K2	2017-2018
	R	What do you understand by critical section? Discuss bakery algorithm. Also show how it satisfies the requirements of a mechanism to control access to critical section	K2	2014-2015
	S	Explain the following terms briefly: (i) Dekker's Solution (ii) Busy Waiting	K2	2014-2015
	T	A barber shop consists of a waiting room with n chairs and a barber room with one barber chair. If there are no customers to be served, the barber goes to sleep. If a customer enters the barbershop and all chairs are occupied, then the customer leaves the shop. If the barber is busy but chairs are available, then the customer sits in one of the free chairs. If the barber is asleep, the customer wakes up the barber. Write an algorithm for the above synchronization problem using semaphores.	K3	2014-2015
	U	What is Bounded Buffer problem? Discuss briefly with the help of Producer and Consumer Process.	K2	2014-2015
	V	Explain in detail about the Mutual Exclusion and Critical Section Problem.	K2	[2021-22]
	W	Explore the concept of deadlock and how it can occur in concurrent systems, discussing prevention and avoidance techniques?		2023-24
	X	Compare and contrast Dekker's and Peterson's algorithms for mutual exclusion, considering their advantages and limitation?		2023-24
	Y	Explain the challenges of achieving mutual exclusion in concurrent programming?		2023-24
	Z	Two semaphores S1 and S2 are both initialized to 1. Process P1 executes wait(S1); wait(S2); and P2 executes wait(S2); wait(S1);. Explain how a deadlock may occur and under what sequence.		2024-25

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UIT	Branch-CSE/AIML/DS	Even Semester 2025-26	Semester: 4th	
	Subject: Operating System		Subject Code: BCS401	
Unit – 3 (Scheduling Concepts)			CO-3	

		Section-A (Short answer type questions)	**Blooms Taxonomy Level**	**Session**
1	A	What are the differences between user level thread and kernel supported thread?	K1	2017-18, 2014-15
	B	Discuss the usage of wait-for graph	K2	2017-18, 2016-17
	C	Discuss the performance criteria for CPU Scheduling	K2	2018-19, 2021-22
	D	What do we need Scheduling?	K1	[2021-22]
	E	What do you mean by the safe state and an unsafe state?	K1	[2021-22]
	F	Explain internal and external fragmentation.	K2	[2022-23
	G	Distinguish between physical and logical address space of a process.	K2	[2022-23]
	H	Define a process state?		2023-24
	I	List three components stored in a PCB		2024-25

Section-B (Long answer type questions)

2	A	Explain multi-threading, illustrate various thread model.	K2	[2014-15]
	B	What is deadlock Explain the various characteristics of a deadlock in detail. Discuss deadlock prevention schemes.	K2	[2017-18]
	C	Draw the state diagram of a process and label various transition. Explain the need of process suspension	K2	2014-15, 2016-17
	D	What is thread? How thread is different from a process?	K2	[2018-19]
	E	Define various types of queues with the help of queuing diagram.	K1	[2014-15]
	F	Draw and explain the Process control block (PCB) with all its components.	K2	[2016-17, 2018-19, [2021-22, 2022-23, 2023-24
	G	Explain various thread models with their relative advantages and disadvantages.	K2	[2015- 16]
	H	Explain the different methods for dealing with the deadlocks.	K2	[2014-15]
	I	Explain short term, medium term and long term scheduling. Describe the differences among them	K2	[2016-17]
	J	Define the following terms: I. Dispatcher II. Dispatch Latency III. Swapping IV. Context Switching	K2	[2015-16]
	K	Explain Banker's algorithm. Why this is used?	K4	[2017-18]
	L	Explain the following scheduling algorithm I. Multilevel feedback queue scheduling II. Multiprocessor Scheduling	K2	[2016-17]
	M	Consider the following snapshot at time T0 of the system and answer the following questions using Banker's Algorithm I. Compute the total number of resources of each type. II. Compute the need matrix III. Is the system in a Safe state? IV. If a request from P1 arrives for (0, 4, 2, 0), can the request granted immediately?	K3	[2015-16]
	N	Consider the set of processes given in the table and draw the Gantt chart for preemptive cases and find out the average waiting time and average turnaround time for following scheduling algorithm.(Note: Larger priority number has higher priority) a. Round Robin b. SRT c. Priority	K3	[2017-18], [2022-23]

Table for 2 M:

Process	Allocation ABCD	Max ABCD	Available ABCD
P0	0012	0012	1520
P1	1000	1750	
P2	1354	2356	
P3	0632	0652	
P4	0014	0656	

<table>
<tr><td></td><td></td><td>
<table>
<tr><td>Process</td><td>Arrival Time</td><td>Execution Time</td><td>Priority</td></tr>
<tr><td>P0</td><td>0</td><td>5</td><td>4</td></tr>
<tr><td>P1</td><td>2</td><td>4</td><td>2</td></tr>
<tr><td>P2</td><td>2</td><td>2</td><td>6</td></tr>
<tr><td>P3</td><td>4</td><td>4</td><td>3</td></tr>
</table>
</td><td></td><td></td></tr>
<tr><td></td><td>O</td><td>List various performance criteria for scheduling algorithms. Five Processes A, B, C, D and E require CPU burst of 3, 5, 2, 5 and 5units respectively. Their arrival times in the system are 0, 1, 3, 9 and 12 respectively. Draw Gantt chart and compute the average turnaround time and average waiting time of these processes for the Shortest Job First (SJF) and Shortest Remaining Time First (SRTF).</td><td>K3</td><td>[2018-19]</td></tr>
<tr><td></td><td>P</td><td>Explain in detail about the Deadlock System model and Deadlock characterization.</td><td>K2</td><td>[2021-22]</td></tr>
<tr><td></td><td>Q</td><td>Consider the given snapshot of a system with five processes (P0,P1,P2,P3,P4) and three resources (A,B,C).
<table>
<tr><td>Process</td><td>Allocation
ABC</td><td>Max
ABC</td><td>Available
ABC</td></tr>
<tr><td>P0</td><td>112</td><td>4 3 3</td><td>210</td></tr>
<tr><td>P1</td><td>212</td><td>3 2 2</td><td></td></tr>
<tr><td>P2</td><td>401</td><td>9 0 2</td><td></td></tr>
<tr><td>P3</td><td>020</td><td>7 5 3</td><td></td></tr>
<tr><td>P4</td><td>112</td><td>11 2 3</td><td></td></tr>
</table>
(i) Calculate the content of Need Matrix. (ii) Apply safety algorithm and check the current system is in safe state or not. (iii) If the request from process P1 arrives for (1,1,0), can the request be granted immediately?</td><td>K3</td><td>[2022-23]</td></tr>
<tr><td></td><td>R</td><td>Explain Deadlock and necessary conditions for deadlock. Also, Explain resource allocation graph with suitable diagram.</td><td>K2</td><td>[2022-23, 2023-24</td></tr>
<tr><td></td><td>S</td><td>Evaluate strategies for deadlock prevention, avoidance, detection, and recovery in concurrent systems, highlighting their effectiveness and trade-offs?</td><td></td><td>2023-24</td></tr>
<tr><td></td><td>T</td><td>Explain the process concept in detail with its states and state transition diagram. Process</td><td></td><td>2024-25</td></tr>
<tr><td></td><td>U</td><td>Consider the set of processes with arrival time (in milliseconds). CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.
<table>
<tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td><td>Priority</td></tr>
<tr><td>P1</td><td>0</td><td>11</td><td>2</td></tr>
<tr><td>P2</td><td>5</td><td>28</td><td>0</td></tr>
<tr><td>P3</td><td>12</td><td>2</td><td>3</td></tr>
<tr><td>P4</td><td>2</td><td>10</td><td>1</td></tr>
<tr><td>P5</td><td>9</td><td>16</td><td>4</td></tr>
</table>
Draw the Gantt chart and find average waiting time using preemptive priority scheduling algorithm.</td><td></td><td>2024-25</td></tr>
<tr><td></td><td>V</td><td>Compare and contrast FCFS, SJF, and Round Robin scheduling algorithms.</td><td></td><td>2024-25</td></tr>
<tr><td></td><td>W</td><td>A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below:
<table>
<tr><td>Process</td><td>Current Allocation</td><td>Maximum Requirement</td></tr>
<tr><td>P1</td><td>3</td><td>7</td></tr>
<tr><td>P2</td><td>1</td><td>6</td></tr>
<tr><td>P3</td><td>3</td><td>5</td></tr>
</table>
Which of the following best describes current state of the system? Explain. 1. Safe, Deadlocked 2. Safe, Not Deadlocked 3. Not Safe, Deadlocked 4. Not Safe, Not deadlocked</td><td></td><td>2024-25</td></tr>
</table>

Question Bank

UIT	Branch-CSE/AIML/DS	Even Semester 2025-26	Semester: 4^{th}	
	Subject: Operating System		Subject Code: BCS401	
Unit – 4 (Memory Management)			CO-4	

		Section-A (Short answer type questions)	Blooms Taxonomy Level	Session
1	A	What is Demand paging?	K1	[2016-17]
	B	Explain Concept of Virtual Memory.	K2	[2016-17]
	C	Difference between External and Internal Fragmentation	K2	[2016-17]
	D	What are the disadvantage of single contiguous memory allocation.	K1	[2017-18]
	E	what is the main function of memory management unit?	K1	[2018-19]
	F	Explain the logical address space and physical address space diagrammatically	K2	[2018-19], 2021-22
	G	Explain in brief about the Multiprogramming with fixed partitions.	K2	[2021-22]
	H	Explain thrashing and locality of reference	K2	[2022-23]
	I	Explain various operations associated with a file.	K2	[2022-23]
	J	Name one piece of process identification information?		2023-24
	K	Define a resident monitor?		2023-24
	L	What is the purpose of a page replacement algorithm?		2024-25
		Section-B (Long answer type questions)		
2	A	Illustrate the page replacement algorithms 3 (i) FIFO (ii) (ii) Optimal page replacement. Using the reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. For a memory with Three frames.	K3	[2018-19]
	B	explain the paging with example. differentiate Paging and Segmentation.	K2	[2018-19]
	C	What do you mean by Belady's anomaly? Which algorithm suffer from Belady's anomaly and how can it rectified?	K3	[2017-18]
	D	Consider the following reference string 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6 How many page fault occur for the FIFO, LRU and Optimal page replacement algorithms, assuming Three and four page in each case and frames are initially empty.	K3	[2015-16], [2017-18]
	E	What is the cause of Thrashing? What steps are taken by the system to eliminate this problem?	K2	[2017-18], [2016-17]
	F	What are the different techniques to remove the fragmentation in case of multiprogramming with fixed partition and variable partitions	K2	[2015-16]
	G	How many page fault would occur for the following reference string for four page frames using LRU and FIFO algorithms. 1,2,3,4,5,5,3,4,1,6,7,8,7,8,9,7,8,9,5, 4,5,4,2.	K3	[2014-15]
	H	Explain segmentation with diagram.	K2	[2014-15]
	I	On a system using paging and segmentation, the virtual address space consists of up to 16 segments where each segment can be up to 216 bytes long. the hardware page each segment into 512 bytes pages how many bits in virtual address specify the following?] (a)Segment Number (b)Page Number (c)Offset within page (d)Entire virtual address	K3	[2014-15]
	J	Explain in detail about the File system protection and security.	K2	[2021-22]

	K	Write short notes on following. i) File system protection and security and ii) Linked File allocation methods	K2	[2021-22]
	L	Explain in detail about the Inter Process Communication models and Schemes.	K2	[2021-22]
	M	Illustrate the following page-replacement algorithms. i) FIFO ii) Optimal Page Replacement Use the reference string 7, 0,1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2,1, 2, 0, 1, 7, 0,1 for a memory with three frames.	K3	[2021-22]
	N	Explain the concept of paging. Also, explain paging hardware support using TLB with suitable diagram	K2	[2022-23]
	O	Explain the concept of demand paging. Consider the given references to the following pages by a program: 0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,3,8,3 How many pages faults will occur if the program has three-page frames available to it and uses: (i) FIFO replacement (ii) LRU replacement (iii) Optimal replacement	K3	[2022-23]
	P	Explain the terms hit ratio and miss ratio. On a simple paged system, associative registers hold the most active page entries and the full-page table is stored in main memory. If references satisfied by associative registers take 100ns, and references through main memory page table takes 180 ns, what must the hit ratio be to achieve an effective access time of 125 ns?	K33	[2022-23]
	Q	Explain the followings: (i) Buffering (ii) Polling (iii) Direct Memory Access (DMA)	K1	[2022-23]
	R	Explain the principles of paging and segmentation as memory management techniques, comparing their suitability for different system architectures?		2023-24
	S	Discuss virtual memory concepts, including address translation, page tables, and demand paging mechanisms?		2023-24
	T	Compare and contrast multiprogramming with fixed partitions and variable partitions?		2023-24
	U	In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page. A process accesses the pages in the following order of page numbers: 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4. If the system has three memory frames that are initially empty, find the number of page faults that will occur during execution of the process.		2024-25
	V	Discuss multiprogramming with variable partitions and how it improves memory utilization.		2024-25
	W	Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?		2024-25

Question Bank

UIT	**Branch-CSE/AIML/DS**	**Even Semester 2025-26**	**Semester: 4th**	
	Subject: Operating System		**Subject Code: BCS401**	
Unit – 5 (I/O devices, and I/O subsystems)			**CO-5**	

		Section-A (Short answer type questions)	**Blooms Taxonomy Level**	**Session**
1	A	Define seek time and rotational latency	K1	[2018-19], [2021-22]
	B	What do you mean by the I/O Buffering?	K1	[2021-22]
	C	Define SCAN and C-SCAN scheduling algorithm.	K1	[2018-19]
	D	Explain tree level directory structure	K2	2023-24
	E	Define an I/O subsystem?		2024-25
		Section-B (Long answer type questions)		
2	A	Describe schemes for defining logical structure of directory	K1	[2014-15]
	B	Discuss disk scheduling algorithm with examples	K2	[2014-15]
	C	What do you mean by cashing, spooling and error handling? Explain in detail. Explain FCFS, SCAN & C-SCAN scheduling with example.	K1	[2016-17]
	D	Write short notes on:- (i) I/O Buffering (ii) Disk Storage and Scheduling	K1	[2017-18]
	E	What are files and explain the access methods for files.	K1	[2018-19], 2022-23
	F	Discuss the Linked, Contiguous and Index file allocation schemes. Which allocation scheme will minimize the amount of space required in directory structure and why?	K2	[2017-18], 2023-24
	G	Write short notes on:- (i)File System Protection and security (ii)Linked File allocation methods	K2	[2017-18]
	H	Explain the following methods:- (i) Bit Vector (ii) Linked list (iii) Grouping (iv) Counting	K2	[2015-16]
	I	What is directory? Explain any two ways to implement the directory.	K2	[2015-16]
	J	Write short notes on:- (i) I/O Buffering (ii) Sequential File (iii) Indexed Fil	K2	[2015-16]
	K	Explain the File organization and Access mechanism.	K2	[2016-17]
	L	A hard disk having 2000 cylinders, numbered from 0 to 1999. The drive is currently serving the request at cylinder 143 and the previous request was at 125. The status of the queue is as follows:- 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 What is the total distance (in cylinder) that the disk arm moves to satisfy the entire pending request for each of the following disk-scheduling algorithm? (i) SSTF (ii) FCFS	K3	[2018-19]
	M	Suppose the moving head disk with 200 tracks is currently serving a request for track 143 and has just finished a request for track 125. If the queue of request is kept in FIFO order 86, 147, 91, 177, 94, 150, 102, 175, 130. What is the total head movement for the following scheduling? (i) FCFS (ii) SSTF (iii) C-SCAN	K3	[2015-16]
	N	Discuss the following terms (i) Access Matrix (ii) Boot Blocks.	K2	[2014-15]
	O	Explain in detail about the Disk storage and Disk scheduling.	K2	[2021-22]
	P	Explain about the concept of File concept. Define in detail about the File organization and access mechanism.	K2	[2021-22]

	Q	A hard disk having 2000 cylinders, numbered from 0 to 1999. the drive is currently serving the request at cylinder 143,and the previous request was at cylinder 125.The status of the queue is as follows 86, 1470, 913, 1774,948,1509,1022,1750,130 What is the total distance (in cylinders) that the disk arm moves to satisfy the entire pending request for each of the following disk scheduling algorithms? (i) SSTF (ii) FCFS	K3	[2021-22]
	R	Explain the term RAID and its characteristics. Also, explain various RAID levels with their advantages and disadvantages.	K2	2022-23, 2024-25
	S	Explain the following terms: (i) Multilevel feedback Queue Scheduling (ii) Fixed portioning vs. variable partitioning	K2	2022-23
	T	Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 49. Calculate the net head movement using: (i) SSTF (ii) SCAN (iii) CSCAN (iv) LOOK	K3	2022-23
	U	Compare and contrast different disk storage technologies, such as HDDs and SSDs?		2023-24
	V	Explain the structure and management of file directories, including hierarchical and flat directory structures?		2023-24
	W	Explain file directories and directory structures in operating systems.		2024-25
	X	Consider a disk queue with requests for I/O to blocks on cylinders 47,38,121,191,87,11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. Calculate the total head movement (in cylinders) required to service these requests.		2024-25
	Y	Discuss RAID 0, RAID 1, and RAID 5 in detail.		